2 Theoretical Underpinnings

2.1 Learning objectives

After completing this chapter, you will be able to:

- State Lotka's, Bradford's, and Zipf's laws and explain the empirical patterns they describe
- Test whether a corpus of publications approximately follows Lotka's inverse-square law
- Construct a Bradford plot to identify core and peripheral journals in a field
- Explain Price's law of exponential growth and the Matthew Effect in science
- Recognise power-law-like distributions in bibliometric data

2.2 Setup

```
library(tidyverse)
library(openalexR)
library(tidytext)
library(glue)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "sci_palette.R"))
```

2.3 Conceptual background

Scientometrics rests on a small number of empirical regularities discovered across decades of studying scholarly communication:

Lotka's Law (Lotka 1926) observes that the number of authors producing n papers is approximately proportional to $1/n^2$. A few prolific authors generate a large share of output, while most researchers publish only one or two papers.

Bradford's Law (Bradford 1934) describes the scattering of articles across journals. For a given topic, a small core of journals contains roughly one-third of all relevant articles; a larger zone contains the next third; and a much larger periphery the final third. Each successive zone contains roughly the same number of articles but geometrically more journals.

Zipf's Law states that the frequency of a word is inversely proportional to its rank. In scientific text, a handful of terms dominate while the vast majority appear rarely.

Price's Law (Solla Price 1963) generalises these observations: the cumulative output of science grows exponentially, doubling roughly every 10–15 years. Price also noted that the square root of all contributors produce half of all contributions.

The **Matthew Effect** (Merton 1968) — named after the Gospel of Matthew ("to those who have, more shall be given") — describes cumulative advantage in science. Well-known researchers attract disproportionate credit and resources, amplifying initial differences in visibility. This mechanism helps explain the skewed distributions observed by Lotka and Price.

2.4 Worked example

We use a corpus of works from the journal *Scientometrics* to demonstrate each law.

2.4.1 Fetching the data

```
works ← fetch_openalex(
  entity = "works",
  primary_location.source.id = "S148561398",
  from_publication_date = "2015-01-01",
  to_publication_date = "2023-12-31",
  options = list(sample = 500, seed = 42)
)
```

2.4.2 Lotka's Law

We unnest authors and count the number of papers per author in this sample.

```
author_prod ← works ▷
  select(id, authorships) ▷
  unnest(authorships, names_sep = "_") ▷
  count(authorships_display_name, name = "n_papers") ▷
  count(n_papers, name = "n_authors") ▷
  arrange(n_papers)

author_prod
```

```
#> # A tibble: 8 × 2
#>   n_papers n_authors
#>      <int>     <int>
#> 1        1       907
#> 2        2       158
#> 3        3        25
#> 4        4         9
#> 5        5         5
#> 6        6         2
#> 7        9         1
#> 8       13         1
```

```r
lotka_c ← author_prod$n_authors[1]

author_prod ▷
  ggplot(aes(x = n_papers, y = n_authors)) +
  geom_point(colour = palette_sci(1), size = 2) +
  geom_line(
    aes(y = lotka_c / n_papers^2),
    linetype = "dashed", colour = "grey40"
  ) +
  scale_x_log10() +
  scale_y_log10() +
  labs(x = "Papers per author", y = "Number of authors") +
  theme_sci()
```

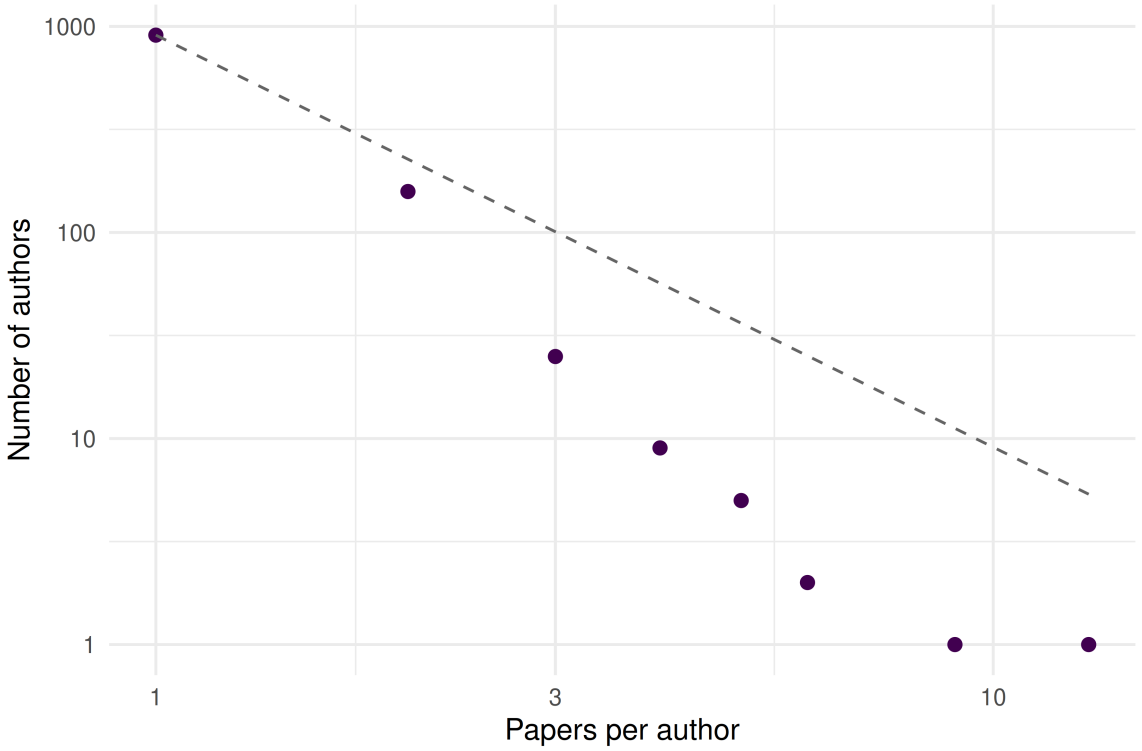


Figure 2.1: Author productivity distribution. The dashed line shows the inverse-square law predicted by Lotka.

The data approximate the inverse-square pattern, though real distributions often deviate from the exact exponent of 2.

2.4.3 Bradford's Law

We count how many articles each journal source contributes (across a broader query).

```r
broad_works ← fetch_openalex(
  entity = "works",
  search = "bibliometrics",
  from_publication_date = "2020-01-01",
  to_publication_date = "2023-12-31",
  options = list(sample = 1000, seed = 42)
)

journal_counts ← broad_works ▷
  filter(!is.na(source_display_name)) ▷
  count(source_display_name, name = "articles", sort = TRUE) ▷
  mutate(
    cum_articles = cumsum(articles),
    rank = row_number()
  )
```

```r
journal_counts ▷
  ggplot(aes(x = rank, y = cum_articles)) +
  geom_line(colour = palette_sci(1), linewidth = 1) +
  scale_x_log10() +
  labs(x = "Journal rank (log scale)", y = "Cumulative articles") +
  theme_sci()
```

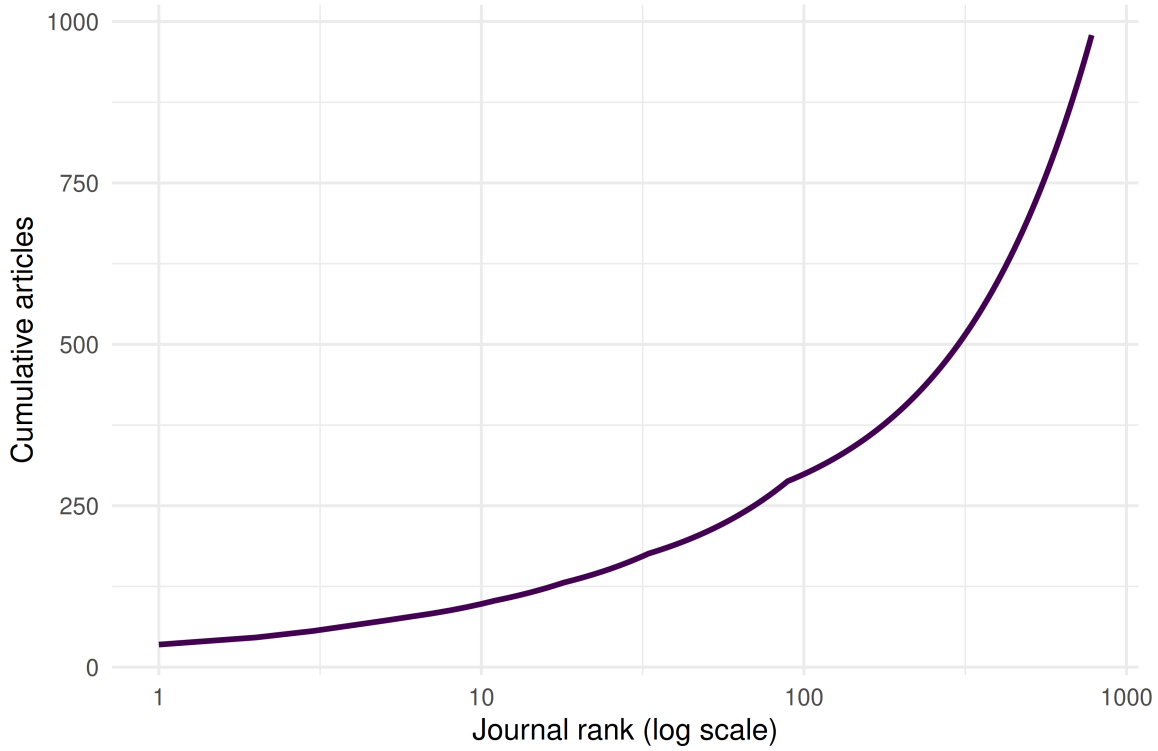


Figure 2.2: Bradford plot: cumulative articles vs. log journal rank.

2.4.4 Zipf's Law

We extract title words and examine their frequency distribution.

```r
word_freq ← works |>
  select(display_name) |>
  unnest_tokens(word, display_name, token = "words") |>
  anti_join(tidytext::get_stopwords(), by = "word") |>
  count(word, sort = TRUE) |>
  mutate(rank = row_number())
```

```r
word_freq |>
  head(200) |>
  ggplot(aes(x = rank, y = n)) +
  geom_point(colour = palette_sci(1), size = 1, alpha = 0.7) +
  scale_x_log10() +
  scale_y_log10() +
  labs(x = "Rank", y = "Frequency") +
  theme_sci()
```

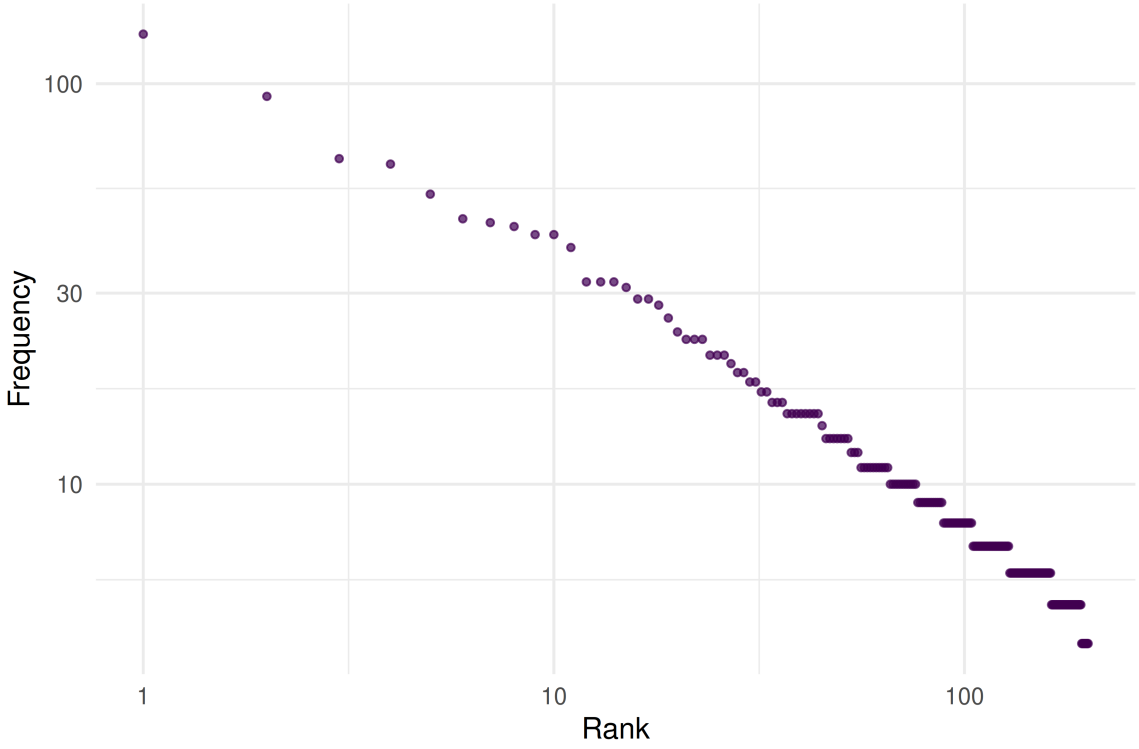


Figure 2.3: Word frequency vs. rank for article titles, approximating Zipf's law.

2.4.5 Price's growth law

```r
works |>
  mutate(year = year(publication_date)) |>
  count(year) |>
  ggplot(aes(x = year, y = n)) +
  geom_col(fill = palette_sci(1)) +
  labs(x = "Year", y = "Works in sample") +
  theme_sci()
```

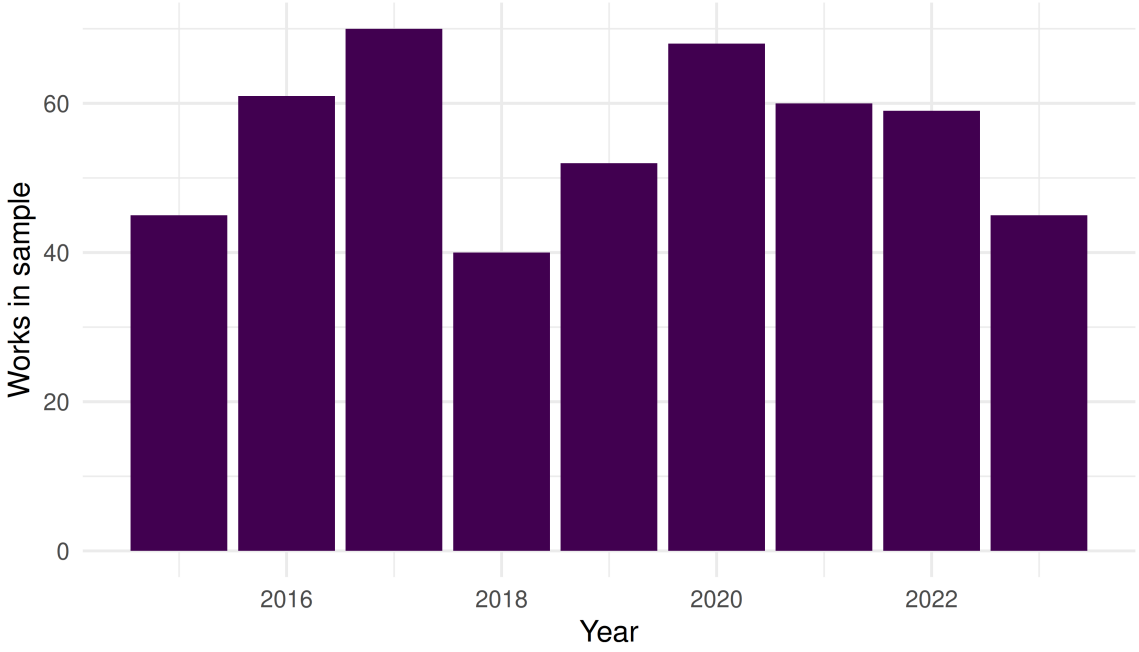


Figure 2.4: Annual publication counts for sampled works, illustrating the growth of bibliometrics literature.

2.5 Diagnostics and interpretation

These laws are *empirical regularities*, not physical laws. They describe approximate patterns:

- Lotka's exponent is often not exactly 2; values between 1.5 and 3.5 are common depending on the field and time window.
- Bradford zones are sensitive to how you define the subject scope.
- Zipf's law holds best for common words; the tail deviates.
- Price's exponential growth eventually saturates — science cannot grow faster than the economy indefinitely.

The value of these laws is not predictive precision but the insight that **skewed distributions are the norm, not the exception**, in scholarly communication.

2.6 Limitations and responsible use

> ## 2.7 Limitations and responsible use
>
> - These laws describe aggregate patterns, not individual behaviour. A researcher publishing one paper is not "less productive" in any evaluative sense — Lotka's law is a statistical observation, not a performance standard.

- The Matthew Effect (Merton 1968) is a mechanism that *amplifies inequality*. Recognising it should prompt caution about using cumulative metrics (like the h-index) that inherit and reinforce this bias.
- Power-law fitting is notoriously difficult. Many distributions that "look" power-law on a log-log plot are better described by lognormal or stretched exponential models. Rigorous fitting requires maximum-likelihood methods and goodness-of-fit tests (Hicks et al. 2015).

2.8 Common pitfalls

2.9 Common pitfalls

- **Eyeballing log-log plots.** A straight line on a log-log plot does not prove a power law. Use formal statistical tests.
- **Confusing description with prescription.** "Most authors publish few papers" does not mean they *should* publish more.
- **Ignoring field differences.** The exponent of Lotka's law and the scattering pattern of Bradford's law vary substantially across disciplines.
- **Assuming stationarity.** These patterns can shift over time as publication norms change (e.g., the rise of mega-journals).

2.10 Exercises

1. **Lotka exponent.** Fit a linear model to the log-log author productivity data. What exponent do you estimate? How does it compare to Lotka's original value of 2?

2. **Bradford zones.** Divide the journal list into three zones of roughly equal article counts. How many journals are in each zone? What is the ratio between successive zone sizes?

3. **Your field's Zipf distribution.** Fetch 500 works in a field of your choice. Extract title words and plot the Zipf distribution. Are the most frequent terms what you would expect?

4. **Matthew Effect simulation.** Write a simple simulation: start 100 authors with 1 paper each. At each time step, an author gains a new paper with probability proportional to their current count. After 50 steps, plot the distribution. Does it resemble Lotka's law?

2.11 Solutions

Solutions are provided in E.

2.12 Further reading

- Lotka (1926) — The original observation on author productivity distributions.
- Bradford (1934) — Journal scattering and the Bradford law.
- Solla Price (1963) — Exponential growth of science and cumulative advantage.
- Merton (1968) — The Matthew Effect: how advantage accumulates in science.
- Garfield (1955) — Citation indexing, the empirical basis for much of scientometrics.
- Waltman (2016) — A modern review contextualising these classical laws within citation impact measurement.

2.13 Session info

```
sessionInfo()
```

```
#> R version 4.6.1 (2026-06-24)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.5 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK
version 3.12.0
#>
#> locale:
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8
#>  [4] LC_COLLATE=C.UTF-8     LC_MONETARY=C.UTF-8    LC_MESSAGES=C.UTF-8
#>  [7] LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
#> [10] LC_TELEPHONE=C         LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] tidytext_0.4.3  glue_1.8.1      openalexR_3.1.2 lubridate_1.9.5
#>  [5] forcats_1.0.1   stringr_1.6.0   dplyr_1.2.1     purrr_1.2.2
#>  [9] readr_2.2.0     tidyr_1.3.2     tibble_3.3.1    ggplot2_4.0.3
#> [13] tidyverse_2.0.0
#>
#> loaded via a namespace (and not attached):
#>  [1] gtable_0.3.6       xfun_0.62          bslib_0.12.0       lattice_0.22-9
#>  [5] tzdb_0.5.0         vctrs_0.7.3        tools_4.6.1        generics_0.1.4
#>  [9] curl_8.1.0         janeaustenr_1.0.0  pkgconfig_2.0.3    tokenizers_0.3.0
#> [13] Matrix_1.7-5       RColorBrewer_1.1-3 S7_0.2.2           lifecycle_1.0.5
#> [17] compiler_4.6.1     farver_2.1.2       SnowballC_0.7.1    htmltools_0.5.9
#> [21] sass_0.4.10        yaml_2.3.12        pillar_1.11.1      jquerylib_0.1.4
#> [25] cachem_1.1.0       viridis_0.6.5      stopwords_2.3      tidyselect_1.2.1
#> [29] digest_0.6.39      stringi_1.8.9      bookdown_0.48      labeling_0.4.3
#> [33] rprojroot_2.1.1    fastmap_1.2.0      grid_4.6.1         here_1.0.2
#> [37] cli_3.6.6          magrittr_2.0.5     utf8_1.2.6         withr_3.0.3
#> [41] scales_1.4.0       timechange_0.4.0   rmarkdown_2.32     httr_1.4.9
#> [45] otel_0.2.0         gridExtra_2.3.1    hms_1.1.4          memoise_2.0.1
#> [49] evaluate_1.0.5     knitr_1.52         viridisLite_0.4.3  rlang_1.3.0
#> [53] Rcpp_1.1.2         downlit_0.4.5      xml2_1.6.0         rstudioapi_0.19.0
#> [57] jsonlite_2.0.0     R6_2.6.1           fs_2.1.0
```